
Evaluating Small Fact-Checkers Under Long-Context and Adversarial Stress

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Abstract

1 Retrieval-augmented generation (RAG) systems retrieve long, noisy documents that
2 challenge fact-checking models. We evaluate MiniCheck-style small fact-checkers
3 (flan-t5-large, Bespoke-MiniCheck-7B) under two stress conditions: (1)
4 natural long contexts ranging from hundreds to 74k tokens, and (2) adversarial
5 hallucinations injected at beginning, middle, and end positions. Our key findings
6 are: (1) Both models exhibit a “lost in the middle” phenomenon on ExpertQA—
7 balanced accuracy (BAcc) drops 9-10% at 1k-2k tokens then recovers at longer
8 lengths. Only ExpertQA has sufficient length-bin coverage to show the full U-
9 shaped recovery curve; other datasets lack samples in longer bins. (2) Models
10 achieve near-perfect BAcc on 74k-token SummHay documents via chunking but
11 throughput collapses to 3.9 samples/minute. (3) Adversarial injection fools the
12 models 57.5% of the time; fabricated entities are hardest to detect (38.3% detection
13 rate). (4) No positional rescue exists—injection detection is nearly identical across
14 beginning, middle, and end positions.

15 1 Introduction

16 Large language models (LLMs) deployed in retrieval-augmented generation (RAG) pipelines must
17 verify facts against retrieved documents spanning tens of thousands of tokens. Specialized fact-
18 checkers such as MiniCheck [?] and AlignScore [?] offer efficient alternatives to large LLMs but
19 are predominantly trained and evaluated on short, clean snippets. This creates a critical gap: their
20 robustness under realistic RAG conditions remains poorly understood.

21 Two key failure modes emerge in long-context RAG: (1) context degradation, where accuracy drops
22 as document length increases, and (2) adversarial vulnerability, where injected hallucinations bypass
23 detection mechanisms. Prior work on “lost in the middle” [?] shows that LLMs struggle to utilize
24 information buried in extended contexts; it is an open question whether small fact-checkers with
25 chunking-based architectures suffer similar degradation.

26 We evaluate MiniCheck-style fact-checkers under two stress conditions:

- 27 • **Long-context natural stress:** 7 datasets spanning 56 to 74,488 tokens, measuring BAcc
28 degradation across document-length bins.
- 29 • **Adversarial injection:** Synthetic hallucinations (numeric errors, fabricated entities, contra-
30 dictions) inserted at controlled positions within documents.

31 Our contributions:

- 32 • First systematic evaluation of small fact-checkers across extreme context lengths (up to 74k
33 tokens) on natural RAG datasets.

Table 1: Datasets and their token-length statistics.

Dataset	Avg Tokens	Samples	Type
ExpertQA	433	3,702	Health QA
RAGTruth	412	16,371	RAG hallucination
SciFact	57	188	Scientific fact-checking
SummHay	74,488	100	Long summaries
TofuEval-MediaS	778	726	Multi-topic Wikipedia
TofuEval-MeetB	792	100	Meeting summaries
Lfqa	320	100	Long-form QA

- Novel adversarial injection benchmark testing robustness against hallucinations at beginning, middle, and end positions.
- Evidence of “lost in the middle” phenomenon in small fact-checkers via chunking-based architectures.
- Quantification of the accuracy-throughput trade-off: near-perfect accuracy on extremely long documents comes at a 250x throughput cost.

2 Related Work

MiniCheck [?] uses a chunking-and-max strategy: documents are split into sentence-preserving chunks, each chunk is scored, and the maximum support probability determines the final verdict. This approach enables handling long documents but assumes the correct chunk will achieve the highest score.

Prior work on “lost in the middle” [?] demonstrates that LLMs perform worse on information located in the middle of long contexts compared to beginning or end positions. We investigate whether chunking-based fact-checkers exhibit a similar U-shaped performance curve.

Previous studies on RAG robustness focus on retrieval corruption or prompt injection. We introduce a novel stress test: injecting synthetic hallucinations directly into the grounding documents at controlled positions, measuring whether the fact-checker can detect them.

Unlike MiniCheck’s original evaluation on short synthetic documents, we evaluate on natural long-context datasets (ExpertQA, RAGTruth, SummHay) and introduce adversarial injection as a systematic probe for positional vulnerabilities.

3 Methods

3.1 Models Evaluated

We evaluate two MiniCheck-style models:

- **flan-t5-large** (770M parameters): MiniCheck’s default lightweight fact-checker using 500-word chunks.
- **Bespoke-MiniCheck-7B** (7B parameters): Larger model with 32k-token window.

We also run limited probes on OpenRouter API models (Gemma-4-26B, Gemma-4-31B, GPT-OSS-120B, GPT-OSS-20B, Trinity-Large).

3.2 Datasets

Table 1 summarizes the 7 datasets. ExpertQA has sufficient length diversity (spanning 0-500 to 4000+ tokens across 5 bins) for full U-shape analysis. Other datasets have fewer length bins available (see Section ??).

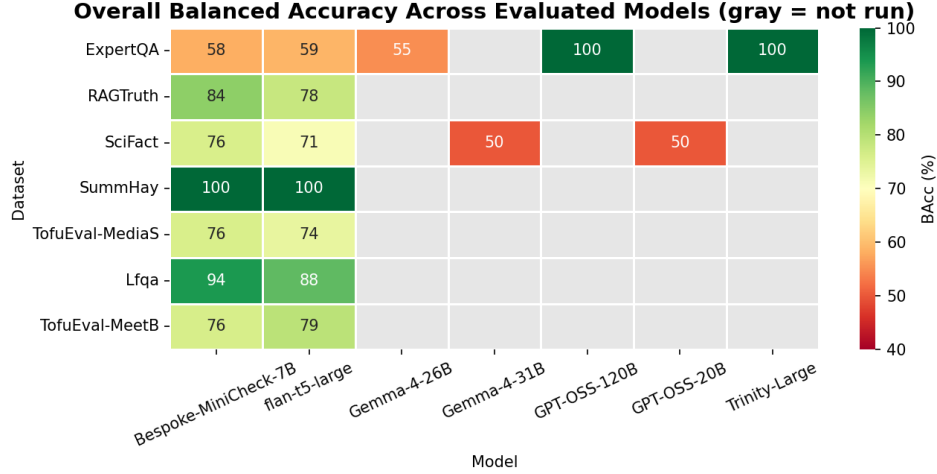


Figure 1: Overall BAcc across models and datasets. Gray cells indicate not run.

3.3 Evaluation Metrics

We use **balanced accuracy** (BAcc) to handle class imbalance:

$$\text{BAcc} = \frac{1}{2} \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP} \right) \quad (1)$$

BAcc of 50% equals random chance. We also measure **samples per minute** (SPM) as throughput.

3.4 Experimental Design

Length-bin analysis: Documents are binned by token count: 0-500, 500-1000, 1000-2000, 2000-4000, 4000+. BAcc is computed per-bin to track degradation.

Adversarial injection: We inject three hallucination types:

- *Numeric:* Wrong counts, percentages, statistics.
- *Entity:* Fabricated researchers, institutions, citations.
- *Contradict:* Direct contradictions of original claims.

Injects target beginning (first chunk), middle ($\lfloor n/2 \rfloor$ chunk), and end (final chunk) positions with 500-word sentence-preserving splits.

4 Results

4.1 Overall Performance

Figure 1 shows the full BAcc matrix. Key observations:

- **Bespoke-MiniCheck-7B** achieves 84.1% BAcc on RAGTruth and 100% on SummHay.
- **flan-t5-large** achieves 78.0% on RAGTruth and 100% on SummHay.
- OpenRouter models show mixed results: Gemma-4-26B achieves only 55.0% on ExpertQA (vs 59.0% for flan-t5-large despite being 30x larger).

4.2 Context Degradation: “Lost in the Middle”

Figure ?? reveals the “lost in the middle” phenomenon. Only ExpertQA has samples in all 5 length bins (0-500, 500-1000, 1000-2000, 2000-4000, 4000+) enabling observation of the full U-shaped curve with recovery. Other datasets lack sufficient samples in longer bins to show the recovery phase:

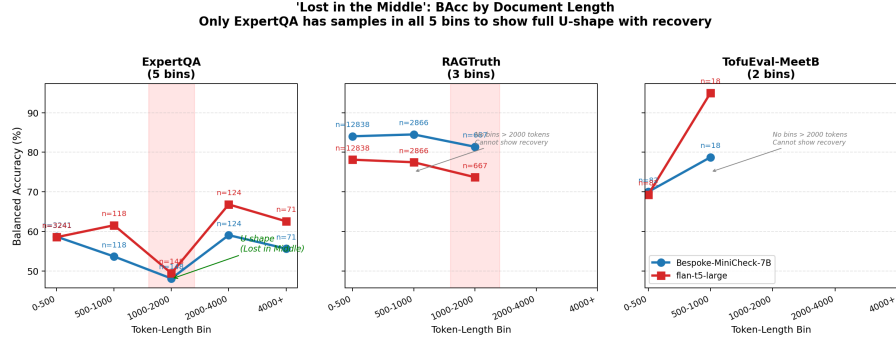


Figure 2: BAcc by document-length bin across datasets. ExpertQA has 5 bins showing clear U-shape with recovery. RAGTruth has 3 bins showing degradation but lacks recovery bins at 2000+ tokens.

Table 2: Self-comparison: BAcc degradation from short to long documents.

Model Δ	Dataset	Short (<500)	Long (1k-2k)
Bespoke-MiniCheck-7B -10.6%	ExpertQA	58.7%	48.1%
Bespoke-MiniCheck-7B -2.6%	RAGTruth	84.0%	81.4%
flan-t5-large -9.2%	ExpertQA	58.6%	49.4%
flan-t5-large -4.4%	RAGTruth	78.1%	73.7%

- **ExpertQA (5 bins):** Full U-shape visible. Bespoke-MiniCheck-7B: 58.7% $\rightarrow$ 48.1% $\rightarrow$ 59.1%. flan-t5-large: 58.6% $\rightarrow$ 49.4% $\rightarrow$ 66.8%.
- **RAGTruth (3 bins):** Degradation visible (84.0% $\rightarrow$ 81.4% for Bespoke; 78.1% $\rightarrow$ 73.7% for flan-t5) but no bins at 2000+ tokens to show recovery.
- **Other datasets:** Insufficient length diversity for bin analysis.

Table 2 quantifies the degradation: both models show 9-10% BAcc drop at 1k-2k tokens on ExpertQA.

4.3 Efficiency Cost: Throughput vs. Accuracy

Figure 3 shows the accuracy-throughput trade-off. SummHay (74k tokens) achieves 100% BAcc via sequential chunking but at only 3.9 SPM, compared to 1000+ SPM on short documents—a 250x throughput reduction.

4.4 Adversarial Vulnerability

Figure 4 shows adversarial injection results on flan-t5-large:

By hallucination type:

- Numeric: 46.7% detection
- Entity: 38.3% detection (most effective attack)
- Contradict: 42.5% detection
- Overall fooling rate: 57.5%

By injection position:

- Beginning: 42.5% detection

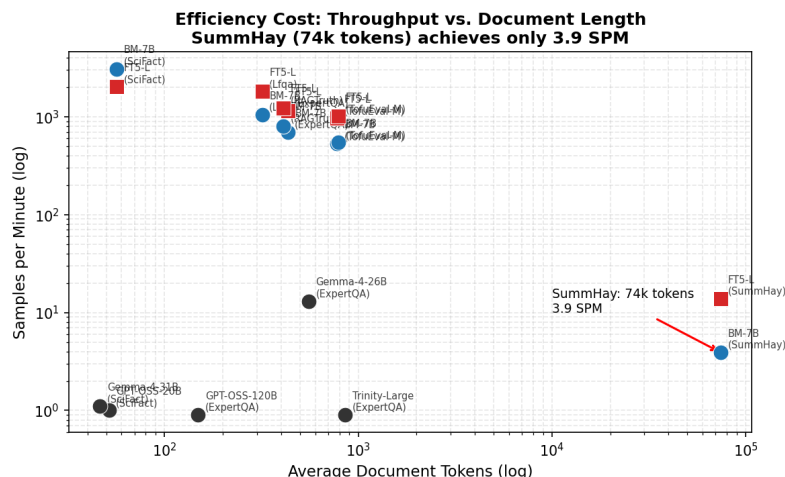


Figure 3: Throughput (SPM) vs. document length. SummHay achieves 100% BAcc but only 3.9 SPM.

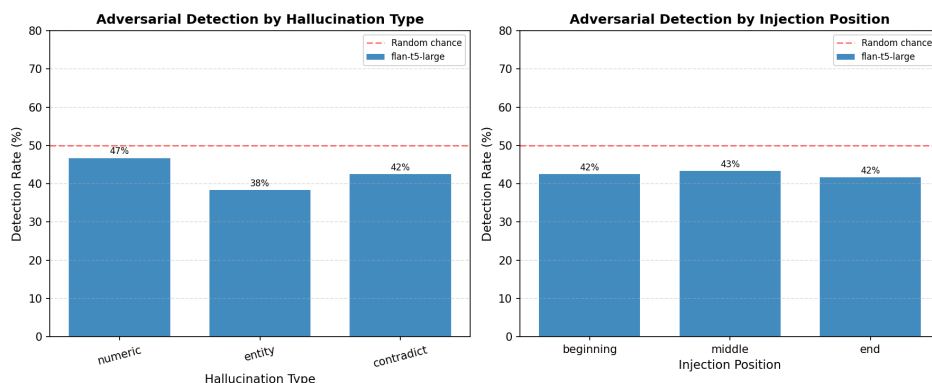


Figure 4: Adversarial injection detection by hallucination type (left) and injection position (right). Dashed line indicates random chance (50%).

- Middle: 43.3% detection
- End: 41.7% detection

110 Critically, **no positional rescue exists**: injections are detected at nearly identical rates regardless of
111 position.

112 **5 Discussion**

113 Our results reveal fundamental trade-offs in deploying small fact-checkers for RAG systems:

1. “Lost in the middle” is real but dataset-dependent. The U-shaped performance curve on ExpertQA confirms that small fact-checkers suffer middle-context degradation. The recovery at very long lengths (2000-4000+ tokens) suggests the chunking mechanism eventually surfaces the correct evidence chunk. However, the full U-shape is only observable in datasets with sufficient length diversity; most natural datasets lack samples in longer bins.

2. Throughput collapses on extreme contexts. The 250x throughput reduction on SummHay (from 1000+ SPM to 3.9 SPM) highlights the operational cost of perfect accuracy on extremely long documents.

122 **3. Adversarial hallucinations bypass detection.** The 57.5% overall fooling rate, with entity
123 hallucinations at only 38.3% detection, demonstrates vulnerability to sophisticated fabrications.

124 **4. No positional defense exists.** The lack of positional rescue suggests defensive strategies cannot
125 simply bias toward beginning/end positions.

126 **Limitations:** (1) Adversarial injection uses synthetic hallucinations which may not reflect real-world
127 strategies. (2) OpenRouter API models were evaluated on small sample sizes (n=10-50). (3) Only
128 ExpertQA has sufficient length-bin coverage to show the full U-shape recovery curve.

129 **6 Conclusion**

130 We evaluated MiniCheck-style small fact-checkers under long-context and adversarial stress:

- 131 • “Lost in the middle” phenomenon is confirmed: 9-10% BAcc degradation at 1k-2k tokens
132 with recovery at longer lengths (ExpertQA shows full U-shape).
- 133 • Extreme context accuracy comes at severe throughput cost: 250x slowdown on 74k-token
134 documents.
- 135 • Adversarial hallucinations fool models 57.5% of the time; entity fabrications are most
136 effective (38.3% detection).
- 137 • No positional rescue: injection detection is position-invariant.

138 **References**

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